

MACROS:

#define VAR1 runs the 1st variations of string checks.

#define VAR1 runs the 2nd variations of string checks which is the staff variation.

Hashing:

The **Jenkins hash functions** are a collection of (non-[cryptographic](#)) [hash functions](#) for multi-[byte](#) keys designed by [Bob Jenkins](#). The first one was formally published in 1997.

one_at_a_time

Jenkins's **one_at_a_time** hash is adapted here from a WWW page by Bob Jenkins,[\[1\]](#) which is an expanded version of his [Dr. Dobbs's](#) article.[\[2\]](#) It was originally created to fulfill certain requirements described by Colin Plumb, a cryptographer.

Each of the 24 rows corresponds to a single bit in the 3-byte input key, and each of the 32 columns corresponds to a bit in the output hash. Colors are chosen by how well the input key bit affects the given output hash bit: a green square indicates good mixing behavior, a yellow square weak mixing behavior, and red would indicate no mixing. Only a few bits in the last byte of the input key are weakly mixed to a minority of bits in the output hash.

Screenshots:

```
Suite: HashTable
  Test: HashTable Insert & Get ...passed
  Test: HashTable Get ...passed
  Test: stringEquals ...passed
  Test: stringHash ...passed
Suite: Task 3
  Test: Empty dictionary ...passed
  Test: Simple ...passed
  Test: Simple without newline ...passed
  Test: Alphanumeric key ...passed
  Test: Multiple entries ...passed
  Test: Non-alphanumeric value ...passed
  Test: Tabs and spaces ...passed

Run Summary:
      Type  Total    Ran Passed Failed Inactive
      suites      2      2    n/a      0      0
      tests     11     11     11      0      0
  asserts    1623   1623   1623      0    n/a

Elapsed time = 0.001 seconds
make[1]: Leaving directory '/home/ats/Desktop/C_Phase/projectC'
make clean
make[1]: Entering directory '/home/ats/Desktop/C_Phase/projectC'
rm -f ./build/*.o philphix unittest
make[1]: Leaving directory '/home/ats/Desktop/C_Phase/projectC'
```

```
touch custom_tests/binary/test_binary.out
rm custom_tests/binary/test_binary.out
=====Running Program.=====
cat custom_tests/binary/test_binary.in | ./philphix custom_tests/binary/test_binary.dict > custom_tests/binary/test_binary.out
Creating hashtable
Loading dictionary custom_tests/binary/test_binary.dict
Dictionary loaded
Processing stdin
=====Program Finished!=====

Difference between test output and reference output
-----
-----None!-----

touch custom_tests/long_word_input/test_long_word_input.out
rm custom_tests/long_word_input/test_long_word_input.out
=====Running Program.=====
cat custom_tests/long_word_input/test_long_word_input.in | ./philphix custom_tests/long_word_input/test_long_word_input.dict > custom_tests/long_word_input/test_long_word_input.out
Creating hashtable
Loading dictionary custom_tests/long_word_input/test_long_word_input.dict
Dictionary loaded
Processing stdin
=====Program Finished!=====

Difference between test output and reference output
-----
-----None!-----

touch custom_tests/long_word_dict/test_long_word_dict.out
rm custom_tests/long_word_dict/test_long_word_dict.out
=====Running Program.=====
cat custom_tests/long_word_dict/test_long_word_dict.in | ./philphix custom_tests/long_word_dict/test_long_word_dict.dict > custom_tests/long_word_dict/test_long_word_dict.out
Creating hashtable
Loading dictionary custom_tests/long_word_dict/test_long_word_dict.dict
Dictionary loaded
Processing stdin
=====Program Finished!=====

Difference between test output and reference output
-----
-----None!-----

You have passed all of the tests!
```

```
None!

touch tests/binary/test_binary.out
rm tests/binary/test_binary.out
=====Running Program.=====
cat tests/binary/test_binary.in | ./philphix tests/binary/test_binary.dict > tests/binary/test_binary.out
Creating hashtable
Loading dictionary tests/binary/test_binary.dict
Dictionary loaded
Processing stdin
=====Program Finished!=====

Difference between test output and reference output
-----None!-----

touch tests/long_word_input/test_long_word_input.out
rm tests/long_word_input/test_long_word_input.out
=====Running Program.=====
cat tests/long_word_input/test_long_word_input.in | ./philphix tests/long_word_input/test_long_word_input.dict > tests/long_word_input/test_long_word_input.out
Creating hashtable
Loading dictionary tests/long_word_input/test_long_word_input.dict
Dictionary loaded
Processing stdin
=====Program Finished!=====

Difference between test output and reference output
-----None!-----

touch tests/long_word_dict/test_long_word_dict.out
rm tests/long_word_dict/test_long_word_dict.out
=====Running Program.=====
cat tests/long_word_dict/test_long_word_dict.in | ./philphix tests/long_word_dict/test_long_word_dict.dict > tests/long_word_dict/test_long_word_dict.out
Creating hashtable
Loading dictionary tests/long_word_dict/test_long_word_dict.dict
Dictionary loaded
Processing stdin
=====Program Finished!=====

Difference between test output and reference output
-----None!-----

You have passed all of the tests!
```